

- 1 (a) (i) $Q = mc\Delta\theta$ C1
 $2300 = 0.75 \times c \times (100 - 20) / 120$...(if uses ± 273 , then -2)..... C1
 $c = 4600 \text{ J kg}^{-1} \text{ K}^{-1}$...(allow 1 sf)..... A1
- (ii) $Q = mL$ C1
 $2300 = (0.375 / 420) \times L$
 $L = 2.6 \times 10^6 \text{ J kg}^{-1}$...(allow 1 sf)..... A1 [5]
- (b) e.g. heat losses, power not constant etc M1
(do not allow if related to s.h.c., rather than l.h.c.)
effect on value for L A1 [2]